

Lesson 8

Troubleshooting OS, Application, and Network Configurations

Topic 8A

Troubleshoot an OS and Applications

Exam Objectives Covered

- 4.4 Given a scenario, troubleshoot common OS and software problems

General Categories

- Hardware
- OS
- Applications
- Misconfigurations
- Patching
- User error

Common Problems

- Unable to log on
- Unable to access resources or files
- System file corruption
- End of life
- Slow performance
- Cannot write to system logs
- Service failures
- System or application hanging or freezing
- Cannot install OS or applications
- Patch update failure

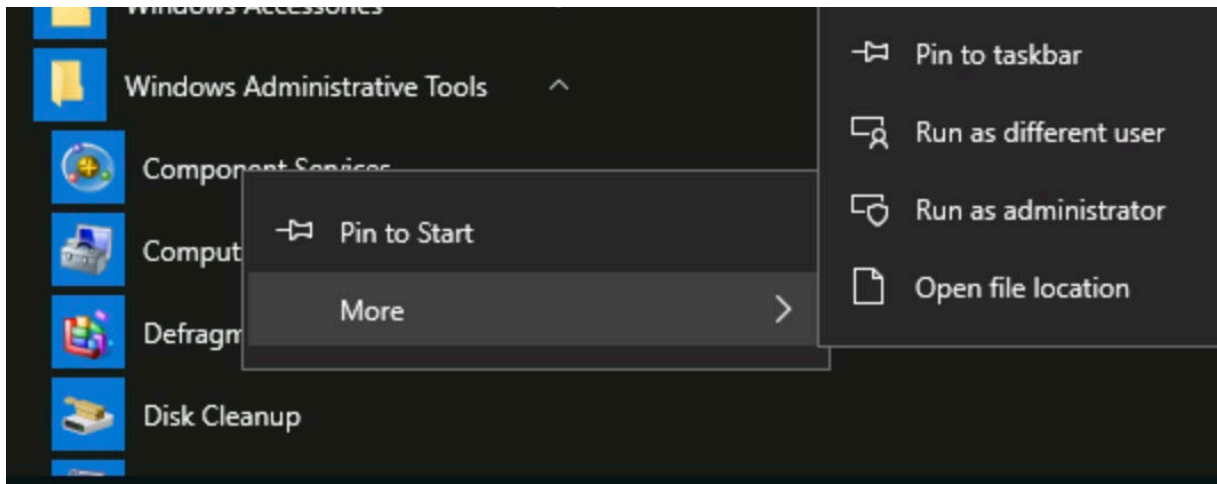
Privilege Escalation

- root
- Administrator
- Domain administrator

```
[root@localhost ~]# whoami  
root  
[root@localhost ~]# █
```

Linux root user.

Privilege Escalation in Windows



The Windows Runas graphical interface. (Screenshot courtesy of Microsoft.)

```
C:\Users\Administrator>runas /user:administrator cmd.exe
Enter the password for administrator: _
```

The Windows runas CLI.

Privilege Escalation in Linux

- Change login credentials

```
[root@localhost ~]# su - devuser  
[devuser@localhost ~]$ █
```

Root user switches to the devuser by using the su command.

Privilege Escalation in Linux (2)

- Delegate privileges

```
[admin@localhost ~]$ useradd testuser
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
[admin@localhost ~]$ sudo useradd testuser
```

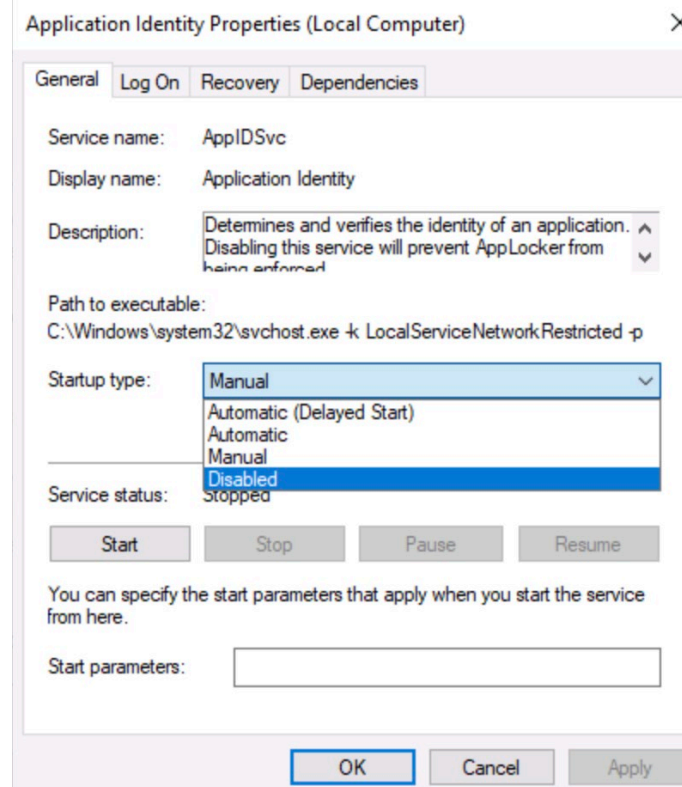
We trust you have received the usual lecture from the local System Administrator. It usually boils down to these three things:

- #1) Respect the privacy of others.
- #2) Think before you type.
- #3) With great power comes great responsibility.

```
[sudo] password for admin:
[admin@localhost ~]$ █
```

Root user switches to the devuser by using the su command.

Services



Configuring a Windows Service startup option. (Screenshot courtesy of Microsoft.)

Services (slide 2)

```
[root@localhost ~]# systemctl start sshd
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; disabled; vendor preset: disabled)
   Active: active (running) since Sun 2021-01-24 08:41:06 MST; 12s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 5884 (sshd)
    Tasks: 1 (limit: 4667)
   Memory: 2.0M
      CPU: 9ms
   CGroup: /system.slice/sshd.service
           └─5884 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Jan 24 08:41:06 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Jan 24 08:41:06 localhost.localdomain sshd[5884]: Server listening on 0.0.0.0 port 22.
Jan 24 08:41:06 localhost.localdomain sshd[5884]: Server listening on :: port 22.
Jan 24 08:41:06 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
[root@localhost ~]# █
```

Starting a service and checking its status in Linux.

Common Categories

- Configuration Manager
- Microsoft Group Policy
- Puppet
- Chef
- Ansible

Configure the Firewall



The Windows Firewall console. (Screenshot courtesy of Microsoft.)

Configure the Firewall (slide 2)

```
[root@localhost ~]# firewall-cmd --zone=public --add-service=http
success
[root@localhost ~]# firewall-cmd --list-all --zone=public
public
  target: default
  icmp-block-inversion: no
  interfaces:
  sources:
  services: dhcpv6-client ftp http mdns ssh
  ports:
  protocols:
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@localhost ~]# █
```

Add the http service to the Linux firewall.

Manage Patches

- Patching
 - Identify
 - Download and deploy to test systems
 - Verify
 - Deploy to remaining systems
- Scheduled reboots

Manage Software Packages

- .msi
- Red Hat
 - RPM
 - YUM

```
[root@localhost ~]# rpm -ivh nmap-7.91-1.x86_64.rpm
Verifying... ##### [100%]
Preparing... ##### [100%]
Updating / installing...
  1:nmap-2:7.91-1 ##### [100%]
[root@localhost ~]# █
```

Use the rpm command to install a software package on Linux.

- Debian
 - dpkg
 - APT

Manage Time Services

- Network Time Protocol (NTP)

```
[root@localhost ~]# cat /etc/chrony.conf
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (https://www.pool.ntp.org/join.html).
pool 2.fedora.pool.ntp.org iburst
```

The Linux NTP /etc/chrony.conf configuration file.

Configure Recovery

- Safe mode--Windows
- Single-user mode—Linux
- Reload OS
- Snapshots

Review Activity: OS and Applications



Review Activity

Topic 8B

Troubleshoot Network Configurations

Exam Objectives Covered

- 4.5 Given a scenario, troubleshoot network connectivity issues

External Issues

- No Internet connectivity
 - Misconfigurations
 - Failures
 - Cables

Internal Issues

- Resource unavailable
- Incorrect or no DHCP information
- Ping failures
- Remote subnets unreachable
- Name resolution issues

Network Troubleshooting Tools

- Verify physical connectivity
- Confirm basic IP configurations
- Test connectivity
- Verify network path
- Update IP configurations
- Display current server connections
- Verify name resolution

Review Activity: Network Configurations



Review Activity

Lesson 8

Summary

